

Experiment No.	7
Experiment Title	Dimensionality Reduction and Model Evaluation (With and Without PCA)
Student Name	L.Radesh
Register Number	3122247001047
Faculty	Dr. Poreddy Ajay Kumar Reddy
Submission Date	31 August 2026
Github Repo	https://github.com/RadeshL/Machine-Learning-Laboratory/blob/main/Mlassign7.ipynb

1 Objective

To study the effect of PCA on multiple classifiers in the original and reduced feature spaces using hyperparameter tuning and 5-fold cross-validation.

2 Dataset Description

The uploaded Spambase dataset contains 4601 samples, 57 predictor features and a binary target. Class 0 contains 2788 samples and class 1 contains 1813 samples. No missing values were found. A stratified 80:20 train-test split was used.

3 Preprocessing

All predictors were standardized using training-set statistics. PCA was fitted only on training data for the With-PCA pipeline.

4 PCA Design

PCA retained 95% of the training variance, reducing the feature space from 57 to 48 components (95.12% variance retained).

Table 1: PCA Variance Explained

Setting	Components	Explained Variance	Justification
With-PCA	48	95.12%	Retains at least 95% of standardized training variance.

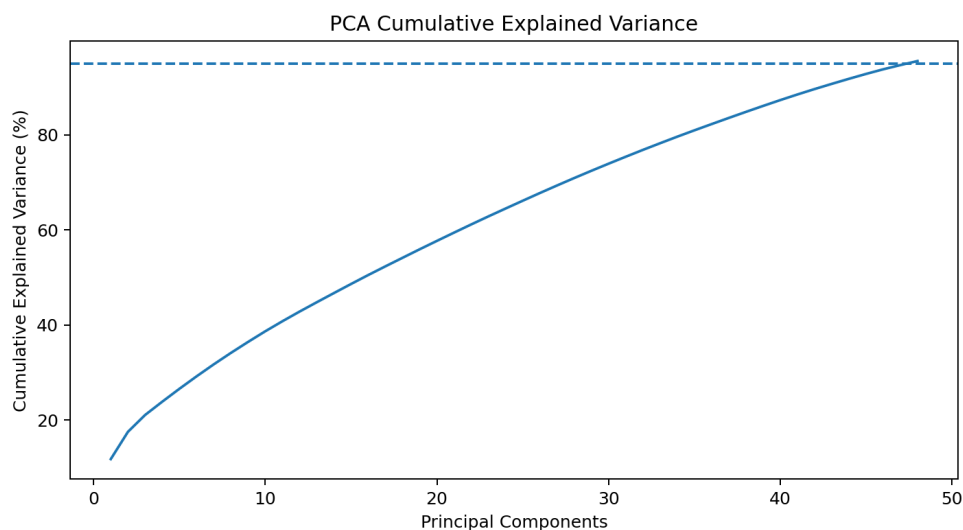


Figure 1: Cumulative explained variance.

5 Hyperparameter Tuning

Candidate configurations were evaluated on a stratified validation split of the training data; the selected configuration was subsequently evaluated with 5-fold cross-validation.

Model	Setting	Values Tried	Selected Setting
SVM	No-PCA	Kernel: linear, RBF; C: 1, 10	rbf
SVM	With-PCA	Kernel: linear, RBF; C: 1, 10	rbf
Naive Bayes	No-PCA	var smoothing: 1e-9, 1e-7	1e-9
Naive Bayes	With-PCA	var smoothing: 1e-9, 1e-7	1e-9
KNN	No-PCA	k: 5, 7; weights: uniform, distance	k7dist
KNN	With-PCA	k: 5, 7; weights: uniform, distance	k7dist
Logistic Regression	No-PCA	C: 1, 10; solver: liblinear	C1
Logistic Regression	With-PCA	C: 1, 10; solver: liblinear	C1
Decision Tree	No-PCA	max depth: 10, 20; min samples split: 2, 5	d20
Decision Tree	With-PCA	max depth: 10, 20; min samples split: 2, 5	d10
Random Forest	No-PCA	n estimators: 50; max depth: None, 15; max features: sqrt	50sqrt
Random Forest	With-PCA	n estimators: 50; max depth: None, 15; max features: sqrt	50d15

AdaBoost	No-PCA	n estimators: 30, 50; learning rate: 1, 0.1	30lr1
AdaBoost	With-PCA	n estimators: 30, 50; learning rate: 1, 0.1	50lr.1
Gradient Boosting	No-PCA	n estimators: 30, 50; max depth: 2, 3; learning rate: 0.1	50d3
Gradient Boosting	With-PCA	n estimators: 30, 50; max depth: 2, 3; learning rate: 0.1	50d3
XGBoost	No-PCA	n estimators: 50; max depth: 3, 5; learning rate: 0.1	50d5
XGBoost	With-PCA	n estimators: 50; max depth: 3, 5; learning rate: 0.1	50d5
Stacking	No-PCA	Base learners: Logistic Regression, KNN, Naive Bayes; meta-learner: Logistic Regression	LR+KNN+NB
Stacking	With-PCA	Base learners: Logistic Regression, KNN, Naive Bayes; meta-learner: Logistic Regression	LR+KNN+NB

6 5-Fold Cross-Validation Results

Weighted F1-score is reported. Each fold cell contains No-PCA / With-PCA.

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Avg No-PCA	Avg With-PCA
SVM	0.9442 / 0.9428	0.9414 / 0.9388	0.9345 / 0.9345	0.9305 / 0.9332	0.9263 / 0.9153	0.9354	0.9354
Naive Bayes	0.8222 / 0.8484	0.8184 / 0.7683	0.8333 / 0.8468	0.8059 / 0.8227	0.8009 / 0.7410	0.8162	0.8162
KNN	0.9211 / 0.9196	0.9198 / 0.9199	0.9238 / 0.9250	0.9275 / 0.9235	0.9116 / 0.9063	0.9208	0.9208
Logistic Regression	0.9373 / 0.9331	0.9252 / 0.9251	0.9300 / 0.9234	0.9073 / 0.9128	0.9168 / 0.9112	0.9233	0.9233
Decision Tree	0.9248 / 0.8812	0.9264 / 0.8903	0.9172 / 0.8906	0.9225 / 0.9023	0.9154 / 0.8709	0.9213	0.9213
Random Forest	0.9605 / 0.9292	0.9537 / 0.9293	0.9564 / 0.9384	0.9468 / 0.9387	0.9467 / 0.9088	0.9529	0.9529
AdaBoost	0.9254 / 0.8696	0.9238 / 0.8777	0.9250 / 0.8850	0.9334 / 0.8742	0.9398 / 0.8640	0.9295	0.9295
Gradient Boosting	0.9387 / 0.9102	0.9372 / 0.9142	0.9413 / 0.9372	0.9372 / 0.9211	0.9385 / 0.8935	0.9386	0.9386
XGBoost	0.9455 / 0.9349	0.9386 / 0.9349	0.9455 / 0.9399	0.9496 / 0.9375	0.9399 / 0.9129	0.9438	0.9438
Stacking	0.9359 / 0.9317	0.9373 / 0.9347	0.9385 / 0.9330	0.9210 / 0.9263	0.9141 / 0.9197	0.9294	0.9294

7 Result Visualization

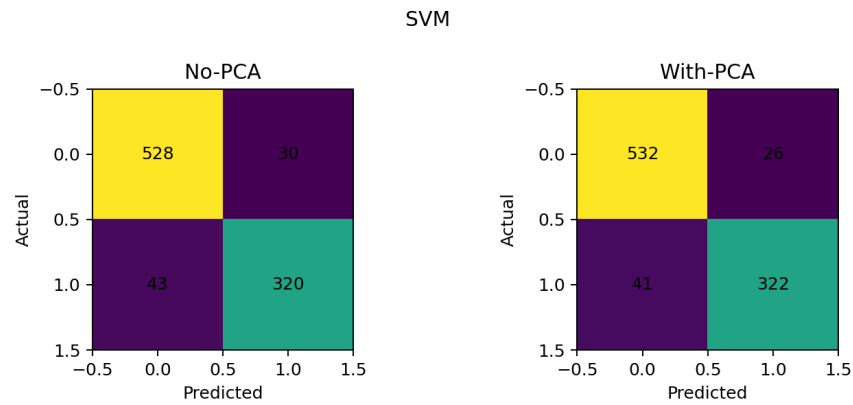


Figure 2: SVM confusion matrices.

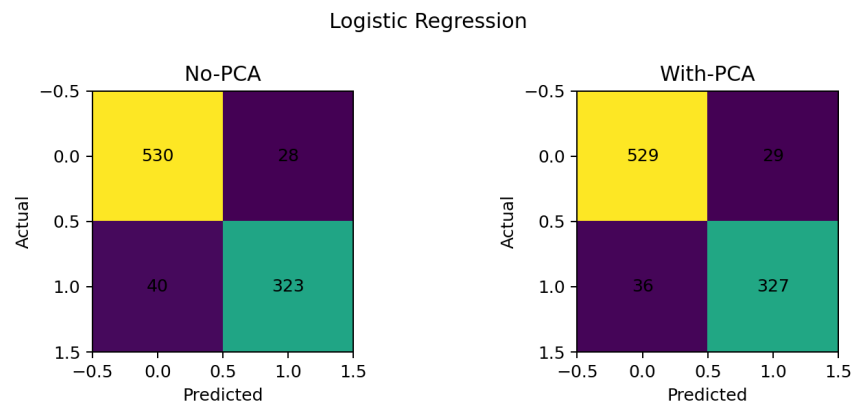


Figure 3: Logistic Regression confusion matrices.

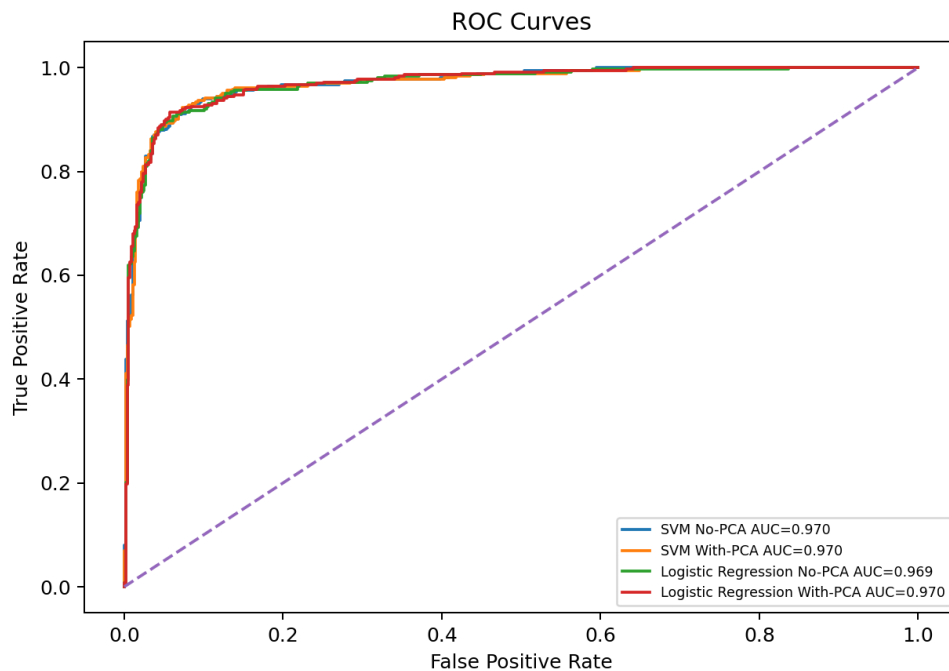


Figure 4: ROC curves.

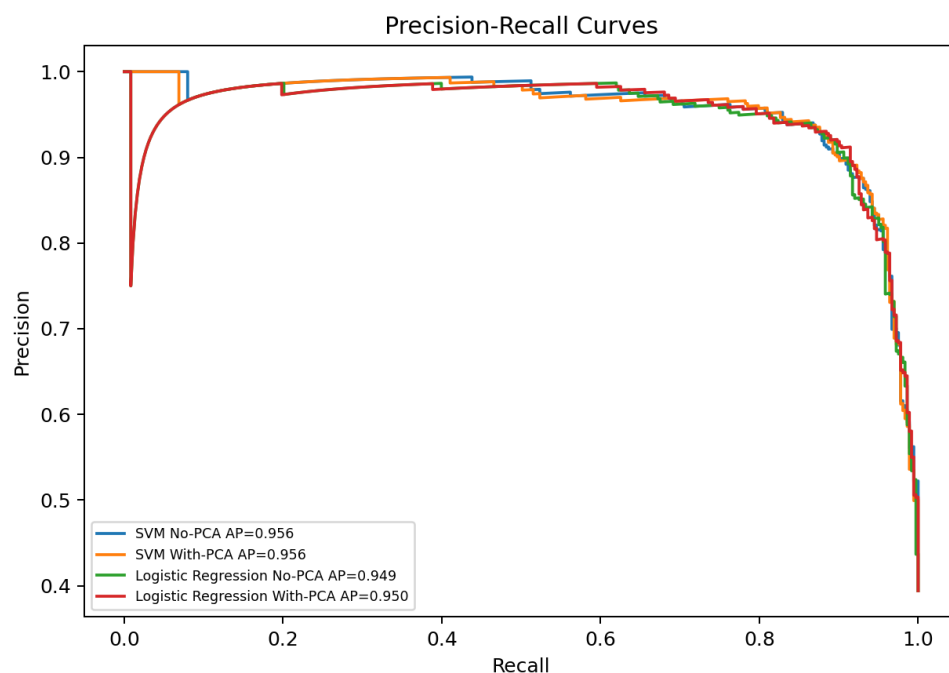


Figure 5: Precision–Recall curves.

8 Observations

The largest positive change in mean 5-fold weighted F1 after PCA was observed for Stacking (-0.0003), while the largest negative change was observed for AdaBoost (-0.0554).

Fold-score standard deviation decreased for 2 of 10 models. PCA therefore had model-dependent effects rather than uniformly improving every classifier.

9 Conclusion

PCA reduced the 57-dimensional feature space to 48 components while retaining 95.12% variance. The measured cross-validation results show how dimensionality reduction affected the individual classifiers.